Are AI coding assistants homogenizing what developers build?

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Milestone: a working historical pipeline and 15 inspected-source project records across three hourly frames.

Our question is open. This pilot tests the measurement process.

Why measure software ideas?

AI suggestions may steer people toward similar projects; easier implementation may also broaden what people build.

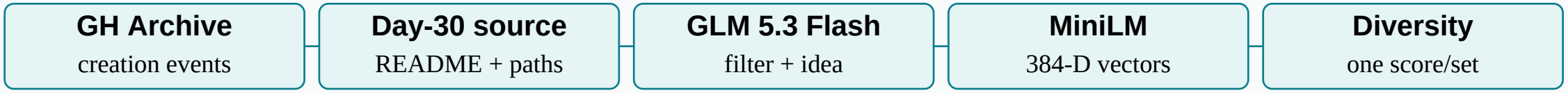
AI-assisted stories became more similar in Doshi & Hauser [1]. Software project choices may behave differently.

Objective: measure changes in new-repository idea diversity around ChatGPT's release [6].

One pipeline, source evidence at every stage

Public creation events → historical source at a fixed repository age → README-grounded idea → embedding → selected-set diversity.

Implemented pilot stages appear in teal. Full-month sampling, human checks and the time-series study remain planned.



PLANNED: whole-month samples -> human validation -> 60-month trend + robustness

Teal = implemented pilot stages. Gray = full-study work; not claimed complete.

What did we actually sample?

January 15, 12:00–12:59 UTC, in 2021, 2023 and 2025 [2]. Seeded order; repository IDs deduplicated.

413 candidates attempted → 15 eligible projects. Every exclusion is logged.

Commits are dated within creation + 30 days, on current-default-branch ancestry [3]. Historical branch identity is not recovered.

These are hourly feasibility samples. Whole-month BigQuery sampling is prepared, not run.

From a README to an idea vector

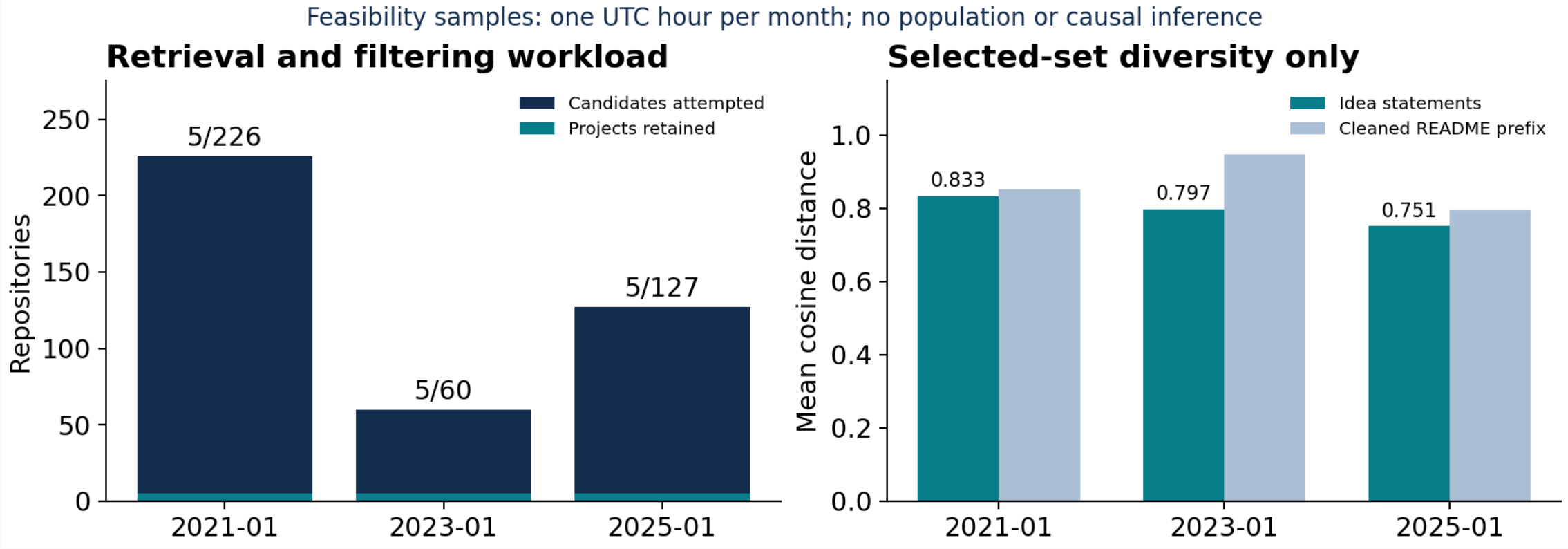
GLM 5.3 Flash: six project labels, an idea ≤80 words, and exact supporting README quotes [7].

Same prompt and settings; bounded historical README input; identity/date redaction is best effort. Reject schema and quotation failures.

MiniLM: 384 dimensions, unit normalization, pinned revision [4]. Mean pairwise cosine distance measures spread.

Estimated API use: $0.0301. A cumulative reservation ledger guards the $5 ceiling.

First descriptive measurements



Checks that already run

Historical commit-window checks; exact quote checks; unit-vector checks; fast score agrees with explicit pairwise distances.

Cleaned README-prefix baseline, leave-one-out ranges, AI-topic exclusion and within-month owner deduplication.

Missing repositories and invalid model outputs are visible results. Passing these checks does not certify summary accuracy.

Human validation is prepared, still pending

15 summary-review rows, 50 pair-rating rows, and a shared rubric in REVIEW.md.

Two independent reviewers: summary purpose/function supported, no invented claims; pair similarity rated from 1 (unrelated) to 5 (same idea).

Then report agreement and similarity correlation. Full study: audit 100 candidate labels, including rejected projects.

No human accuracy or agreement percentage is claimed today.

What we can and cannot conclude

We can run the pipeline, preserve historical links, quantify attrition, and measure selected-set diversity.

We cannot infer an AI effect, population monthly diversity, or a reliable temporal trend from three sparse hourly frames.

Main risks: survival/documentation bias, changed branches/history, model errors, and unobserved changes in who uses GitHub.

Next: whole-month frames, independent review, broader pilot, frozen prompt/sample size, then 60-month analysis [5,8].

Milestone status and next steps

Samuel: working data pipeline; next, full-month frames and historical retrieval checks.

Spencer: ideas, embeddings and descriptive scores; next, several-month pilot and full trend after validation.

Hezeki + all: prepared review guides; next, independent ratings and evidence review before submission.

Oct 28–Nov 8: sampling/measurement pilot. Nov 9–20: collection and validation. Nov 21–final: analysis and report.

References and AI disclosure

[1] Doshi & Hauser, Science Advances (2024). [2] GH Archive. [3] GitHub commit API. [4] MiniLM model card.

[5] Lopez Bernal et al., IJE (2017). [6] OpenAI, Introducing ChatGPT (2022). [7] Prime Inference. [8] Newey & West (1987).

Full linked references are in the writeup. AI_USE.md records prompts, settings and errors. Codex drafted code/materials; GLM produced classifications, ideas and design advice.

Team review and acceptance decisions remain pending. Questions?